

Please check the examination details below before entering your candidate information

Candidate surname

Other names

Pearson Edexcel
International
Advanced Level

Centre Number

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Candidate Number

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Friday 16 October 2020

Afternoon (Time: 1 hour 30 minutes)

Paper Reference **WME01/01**

Mathematics

International Advanced Subsidiary/Advanced Level
Mechanics M1

You must have:

Mathematical Formulae and Statistical Tables (Blue), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$, and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for each question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- If you change your mind about an answer, cross it out and put your new answer and any working underneath.

Turn over ►

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Pearson

Question 1 continued

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Q1

(Total 7 marks)



Question 2 continued

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Question 2 continued

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Q2

(Total 14 marks)



3.

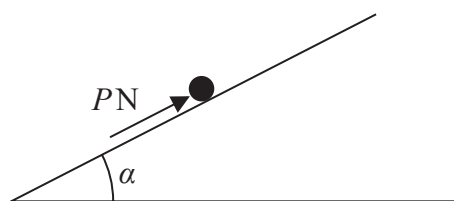


Figure 1

A particle of mass 10 kg is placed on a fixed rough inclined plane. The plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{3}{4}$. The particle is held in equilibrium by a force of magnitude P newtons, which acts up the plane, as shown in Figure 1. The line of action of the force lies in a vertical plane that contains a line of greatest slope of the plane. The coefficient of friction between the particle and the plane is $\frac{1}{2}$.

- (a) Find the normal reaction between the particle and the plane. (2)
- (b) Find the greatest possible value of P . (4)
- (c) Find the least possible value of P . (2)



Question 3 continued

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Question 3 continued

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Question 3 continued

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Q3



Question 4 continued

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Question 4 continued

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Q4



- (d) Find the value of t when P is moving in the direction of the vector $(-5\mathbf{i} + 8\mathbf{j})$ (4)

Question 5 continued

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Question 5 continued

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Question 5 continued

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Q5



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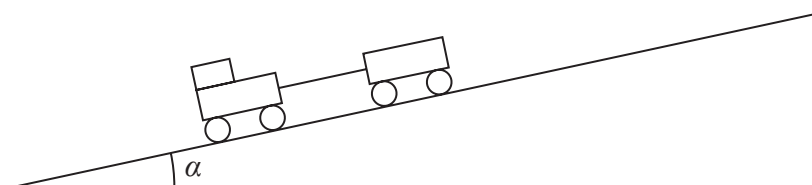


Figure 3

A railway engine of mass 1500 kg is attached to a railway truck of mass 500 kg by a straight rigid coupling. The engine pushes the truck up a straight track, which is inclined to the horizontal at an angle α , where $\sin \alpha = \frac{7}{25}$. The coupling is parallel to the track and parallel to the direction of motion, as shown in Figure 3.

The engine produces a constant driving force of magnitude D newtons. The engine and the truck experience constant resistances to motion, from non-gravitational forces, of magnitude 1200 N and 500 N respectively.

The thrust in the coupling is 2000 N.

The coupling is modelled as a light rod.

- (a) Find the acceleration of the engine and the truck.

(4)

- (b) Find the value of D .

(4)



Question 6 continued

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Question 6 continued

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Question 6 continued

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(Total 8 marks)

Q6



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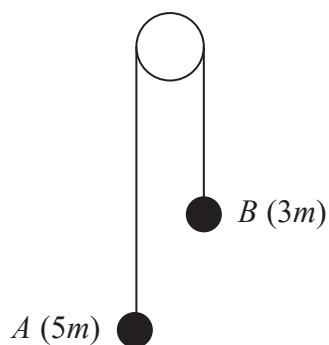


Figure 4

One end of a light inextensible string is attached to a particle A of mass $5m$. The other end of the string is attached to a particle B of mass $3m$. The string passes over a small, smooth, light fixed pulley. Particle A is held at rest with the string taut and the hanging parts of the string vertical, as shown in Figure 4.

Particle A is released.

- (a) Find, in terms of m and g , the magnitude of the force exerted on the pulley by the string while A is falling and before B hits the pulley. (8)
- (b) State how, in your solution to part (a), you have used the fact that the pulley is smooth. (1)



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Question 7 continued

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Question 7 continued

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Question 7 continued

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Q7



Question 8 continued

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Question 8 continued

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Q8

TOTAL FOR PAPER = 75 MARKS

END